

Насколько хорошо ChatGPT-3.5 справляется с семантическими отношениями в новостных текстах?

<https://nicolay-r.github.io>

30 августа 2023

Эволюция моделей машинного обучения

- Мешок слов (доклады ранее)
- Embedding векторы (доклады ранее)
- Машинный перевод → Механизм внимания → Языковые модели

Представление текста: Контекстно-зависимый вектор слова (III)
Установление внимания на нескольких слоях, и в рамках одного языка
(переход от машинного перевода к одному тексту)



Диалоговые системы в индустрии

AI к NPC в демо-версии Unreal-Engine 5. ¹

²

Описание применения ChatGPT для анализов PDF-файлов статей^{3,4}

1 <https://www.youtube.com/watch?v=4sCWf2VGdfc>

2 https://www.youtube.com/results?search_query=this+ai+killed+3+people

3 <https://www.chatpdf.com/>

4 <https://askyourpdf.com/>

Применение ChatGPT к анализу новостей

Применение ChatGPT к анализу новостей

- В презентации рассматриваются новостные тексты *на русском языке*.
- Под **отношениями** понимаются связи двух объектов.
- **Цель:** зная отношения, размеченные вручную, проверить насколько такая разметка совпадает с автоматической.

Построение запроса к ChatGPT

Подсказка (prompt) – универсальный формат запроса к диалоговой системе на естественном языке.

Изначальный текст подсказки:

Для текста “...” существует ли отношение типа X от ОБЪЕКТА_А к ОБЪЕКТУ_В?

Однако, сторонние исследования показывают, что предпочтительнее рассматривать версию на английском языке^[1].

[1] Bowen Zhang, Daijun Ding и Liwen Jing. «How would stance detection techniques evolve after the launch of chatgpt?» В: *arXiv preprint arXiv:2212.14548* (2022).

Анализ тональности^[2]

What is the attitude of the sentence [...] to the target OBJ? Select one from «favor, against or neutral» and explain why

Participant	F1-PN-macro (rank)	F1-PN0-macro (rank)
mtsai	66.67 (1)	74.11 (2)
cookies	66.64 (2)	74.29 (1)
lsanochkin	62.92 (3)	71.20 (3)
ChatGPT **	60.06 (-)	70.79 (-)
antongolubev	59.64 (4)	69.04 (4)

[2] Anton Golubev, Nicolay Rusnachenko и Natalia Loukachevitch. «RuSentNE-2023: Evaluating Entity-Oriented Sentiment Analysis on Russian News Texts». В: *Computational Linguistics and Intellectual Technologies: papers from the Annual conference "Dialogue"* (arxiv:2305.17679. 2023).

Формат проведения эксперимента

Используется большая открытая коллекция новостных текстов на русском языке NEREL^[3] (900 документов)

- Используемый текст подсказки:

For text "... " is the relation of type X from OBJ_A to OBJ_B?

- Версия ChatGPT-3.5-Turbo (даты извлечения июль-август)
- метрика оценки: Точность (Accuracy) число верных примеров / общее число

[3] Natalia Loukachevitch и др. «NEREL: A Russian Dataset with Nested Named Entities, Relations and Events». В: *Proceedings of the International Conference on Recent Advances in Natural Language Processing (RANLP 2021)*. Held Online: INCOMA Ltd., сент. 2021, с. 876—885. URL: <https://aclanthology.org/2021.ranlp-1.100>.

Типы рассматриваемых отношений

Временные: CAUSE_OF_DEATH, DATE_OF_BIRTH, DATE_OF_DEATH, PLACE_OF_DEATH, END_TIME, START_TIME, PLACE_OF_BIRTH, DATE_OF_CREATION, DATE_FOUNDED_IN, DATE_DEFUNCT_IN

Принадлежность/часть: PART_OF, OWNER_OF, MEMBER_OF, CONVICTED_OF, SUBORDINATE_OF, SUBEVENT_OF

Деятельность/участие: WORKS_AS, AWARDED_WITH, PARTICIPANT_IN, ALTERNATIVE_NAME, FOUNDED_BY, ORGANIZES

Местоположение: LOCATED_IN, TAKES_PLACE_IN, MEDICAL_CONDITION, SCHOOLS_ATTENDED, WORKPLACE

Родственные/общие: RELATIVE, SIBLING, SPOUSE, PARENT_OF, KNOWS

формат рассмотрения результатов



Рис.: Разделение результатов на 3 группы: **высокий** уровень согласия, **средний** результат согласия, и **низкий**

Примеры разметок с высокой точностью

Наиболее успешно распознанные связи (70-90%)

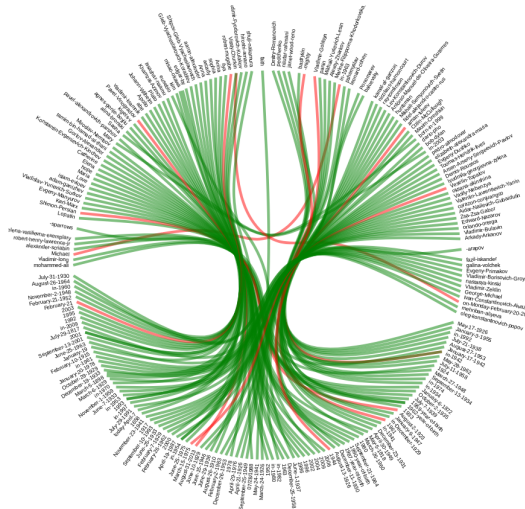
Достаточно хорошо распознаются **временные связи**.

Тип отношения	Точность (%)	Количество
CAUSE_OF_DEATH	93	41
DATE_OF_BIRTH	92	114
DATE_OF_DEATH	86	87
PLACE_OF_DEATH	86	63
END_TIME	86	22
START_TIME	82	38
PLACE_OF_BIRTH	76	97
DATE_OF_CREATION	75	117

DATE_OF_BIRTH (Анализ расхождения результатов)

Основные причины ошибок
(проанализировано 5
примеров):

- ошибки перевода (2/5);
- галлюцинации (2/5);
- расхождение даты контекста (1/5);
- неточность вопроса (1/5);



DATE_OF_BIRTH. Ошибки перевода и исходного текста (I)

For text: 'aman-tuleev 73-years-old (1944 R.) .', is the relation of type DATE_OF_BIRTH from 'aman tuleev' towards '1944'? Answer yes or no, and explain why if no.

ChatGPT

No, the relation of type DATE_OF_BIRTH does not exist from "aman tuleev" towards "1944" in the given text. The phrase "DATE_OF_BIRTH" typically indicates a relationship between a person and their date of birth. In this case, "aman tuleev" is the person, and "1944" is a year mentioned in the text. However, the **text does not explicitly state** that "1944" is the date of birth of Aman Tuleev. It only provides information that he is 73 years old, **and his birth year is mentioned as "1944 R."**

DATE_OF_BIRTH. Ошибки неточного перевода (II)

For text: 'The new crew , which included two Russians — Pavel-Vinogradov And alexander-misurkin ... **Pavel-Vinogradov August-31,-2013 year** will meet in space 60th-anniversary , and for him it is already third space-flight .', is the relation of type DATE_OF_BIRTH from 'Pavel Vinogradov' towards 'August 31, 2013'?

ChatGPT

The relation of type DATE_OF_BIRTH implies that 'Pavel Vinogradov's date of birth is "August 31, 2013." However, in the given text, **there is no mention or indication that "August 31, 2013" is the date of birth of Pavel Vinogradov.** Instead, the text mentions that on August 31, 2013, Pavel Vinogradov will meet in space for his 60th anniversary, indicating that it is the date of a milestone event related to his age or spaceflight experience. Therefore, the relation of type DATE_OF_BIRTH is not applicable in this context.

Связь DATE_OF_BIRTH: Галлюцинация ChatGPT

For text: 'in-**1963** at marina-filippovna And Boris-Moiseevich was-born son **Michael**, which later became entrepreneur and led nc-yukos .', is the relation of type DATE_OF_BIRTH from 'Michael' towards 'in 1963'?

ChatGPT

The text mentions that in 1963, Marina-Filippovna and Boris-Moiseevich **had a son named Michael**. The phrase "in 1963" is **associated with the birth of their son** and not with Michael himself. Therefore, the relation of type DATE_OF_BIRTH does not apply to "in 1963" in relation to Michael. Instead, it indicates the year of Michael's birth.

Связь PLACE_OF_BIRTH и PLACE_OF_DEATH

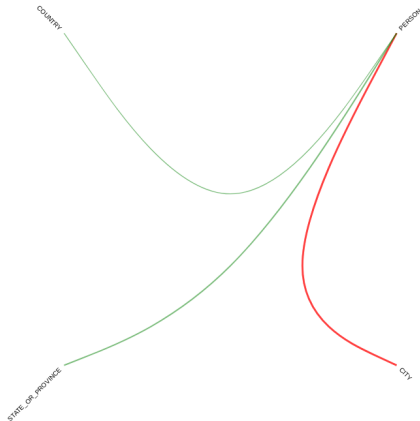


Рис.: Наибольшая часть ошибок возникает в случае, когда связь образуется между городом и личностью (PERSON→CITY); пример на следующем слайде

Связь PLACE_OF_BIRTH: Более детальное пояснение примера со стороны ChatGPT

"For text: '**Maxim-Oreshkin** was-born-in-1982 . He spent his youth in Moscow , **Khovrino** . oreshkin It has ten-year-old experience in the commercial field , having worked in rosbank , vtb-capital , also held the post leading-economist V central-bank and managed to work in government department-director , then Deputy-Minister-of-Finance .'

ChatGPT

The text provides information about 'Maxim Oreshkin's date of birth (in 1982) **but does not explicitly state his place of birth.** It only mentions that he spent his youth in Moscow, specifically in the locality of Khovrino. However, **spending his youth in Khovrino does not necessarily mean that Khovrino is his place of birth.** The relation of type PLACE_OF_BIRTH typically indicates the specific location where an individual was born.

Примеры разметок со средней точностью

50-70% Точность

Общее число отношений значительно больше чем те, что были более точно распознаны.

Тип отношения	Точность (%)	Количество
WORKS_AS	71	3053
LOCATED_IN	71	611
TAKES_PLACE_IN	70	1222
SUBEVENT_OF	68	212
AWARDED_WITH	64	401
ORIGINS_FROM	61	956
OWNER_OF	54	94
WORKPLACE	53	804

Связь WORKS_AS (Анализ результатов расхождений)

Анализ 10 примеров:

- Difference between occupation/post and position (6/10).
- Some cases are ended with the mistakes (2/10).
- Context-related reasoning (1/10).
- Bad translation (1/10).

Вывод: rigorous meaning for the provided examples. In general, model might reasoning from other angles that are slightly close to works as.

Связь WORKS_AS: Рассуждение в рамках контекста

Alice-munro — 110th winner and 13th woman in her nomination . She is a laureate Booker-Prize . Some colleagues call Canadian **writer** "our **Chekhov**" , comparing it with the great Russian writer , known as the author of interesting short stories ...

ChatGPT

The phrase "our Chekhov" is a figurative comparison, highlighting the similarities in their writing styles or themes. However, it does not imply that Anton Chekhov worked as a writer, nor does it establish a direct WORKS_AS relationship between 'Chekhov' and 'writer' **in this context**. Therefore, the relation of WORKS_AS is not applicable in this case.

Связь WORKS_AS: Ошибки рассуждений

'in-March-2016 V Moscow celebrated wedding Said-Gutseriev - son billionaire Mikhail-Gutseriev - And Khadija-Uzhakhova . Among entertaining guests **artists** were Jennifer-Lopez , **sting** And Enrique-Iglesias .'

ChatGPT

No, the relation of type WORKS_AS does not exist from 'sting' towards 'artists'. The text mentions that **Sting was among the entertaining guests** at the wedding, **but it does not specify his role** as an artist in this context. Therefore, **we cannot determine from this text whether Sting was working as an artist** at the wedding.

Разница в понятиях: пост/занятие и позиция (I)

The ex- head-of-the-Bryansk-branch-of-the-Yabloko-party and former contender gubernatorial post on elections October-14 **andrey-ponomarev** appointed **and-about.-Deputy-Governor-of-the-Bryansk-Region-for-Economics**.

ChatGPT (appointed != work)

The phrase "and-about. Deputy Governor of the Bryansk Region for Economics" seems to be a description of the position or title that Andrey Ponomarev **was appointed to**, rather than a specific job role or workplace that he works as. The sentence mentions that he was appointed as the Deputy Governor for Economics, but **it does not provide** a clear indication that he works as "and-about. Deputy Governor of the Bryansk Region for Economics."

Разница в понятиях: пост/занятие и позиция (II)

Miroslav Nemtsov → Chairman of the Lower House of Parliament

For text: 'government-coalition from ods , top-09 and their junior partner - liberal-democratic slide would like to resume government . Therefore , the top ods agreed very quickly and Thursday-20-June put-forward-a-candidate to post prime-minister - **chairman-of-the-lower-house-of-parliament** **Miroslav-Nemtsov** (1952 birth) ...'

ChatGPT

The text mentions that Miroslav Nemtsov is the candidate put forward for **the post of prime minister** and chairman of the lower house of parliament, but **it doesn't explicitly state that he currently works** as the chairman of the lower house of parliament. Therefore, the relation of type WORKS_AS cannot be established in this context.

WORKPLACE и OWNER_OF (Анализ результатов)

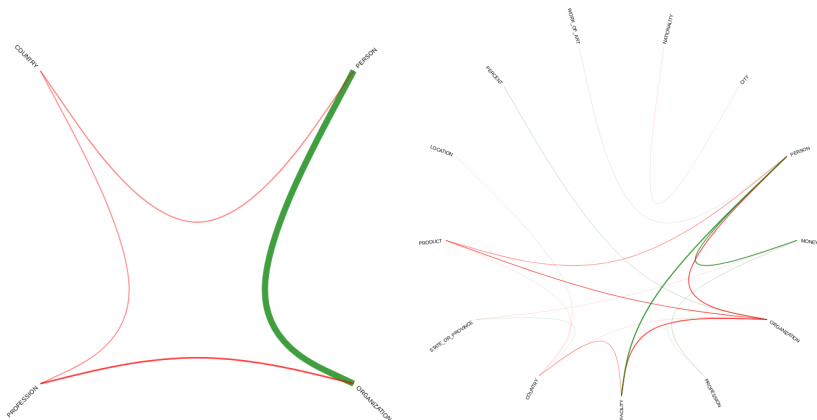


Рис.: Анализ связей между типами WORKPLACE (слева) и OWNER_OF (справа)

Примеры разметок с низкой точностью

Наиболее трудно распознаваемые связи (<40%)

Тип отношения	Точность (%)	Количество
PARTICIPANT_IN	42	2764
HAS_CAUSE	42	481
KNOWS	41	264
EXPENDITURE	41	29
ALTERNATIVE_NAME	40	942
FOUNDED_BY	37	86
ORGANIZES	30	123
CONVICTED_OF	22	286
SUBORDINATE_OF	15	105

Связь FOUNDED_BY

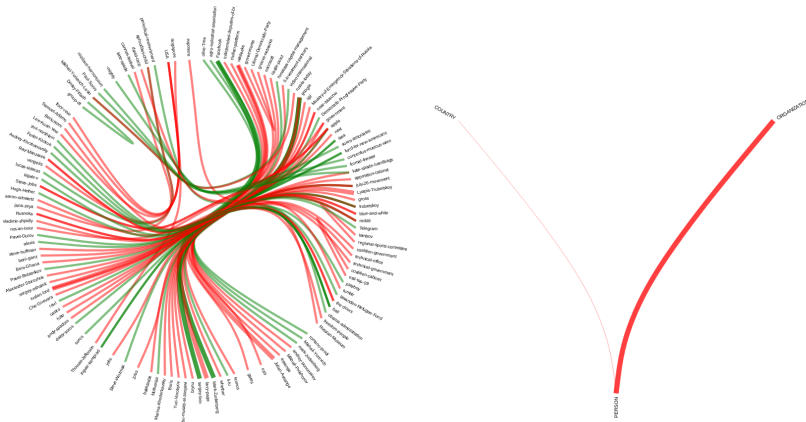


Рис.: Пример иллюстрирует наличие основных ошибок связей между PERSON→ORGANIZATION

Связь SUBORDINATE_OF (требуется уточнение понятия)

'manager mia-farrow said his family " mourns the tragic loss of a beloved husband , father , grandfather , brother and uncle . " "They really appreciate the support of friends during this difficult time ," he added', is the relation of type SUBORDINATE_OF from 'manager' towards 'mia farrow' ?

ChatGPT

The text mentions that "manager Mia Farrow said," which implies that the manager is speaking on behalf of or representing Mia Farrow. However, this does not establish a hierarchical or subordinate relationship between the manager and Mia Farrow. Instead, it suggests that **the manager is speaking in a professional capacity** on behalf of Mia Farrow and her family.

Есть проблема перевода, однако упоминание термина "менеджер" (Manager) не соответствует отношению SUBORDINATE_OF

Связь SUBORDINATE_OF: Недостаток информации (I)

"Head-of-the-Main-Organizational-and-Inspection-Department-of-the-Prosecutor-General's-Office yuri-sindeev left supervisory-authority . Cross on career 45-year-old **general**, nearest assistant Prosecutor-General **yuri-seagull**, put clandestine-casino-cover-up-scandal V suburbs"Is the relation of type SUBORDINATE_OF from "general" towards "yuri seagull"?

ChatGPT

The text mentions that Yuri Seagull is the "nearest assistant Prosecutor-General," but it does not explicitly state that the "general" mentioned in the text is subordinate to Yuri Seagull. **The term "general" could refer to a military rank or a general position without any direct indication of being subordinate to Yuri Seagull.**

В данном случае "nearest assistant" (ближайший помощник) предполагает SUBORDINATE_OF.

Связь SUBORDINATE_OF: Недостаток информации (II)

"concert-director **actresses stas-khristov** told the agency Moscow:" is the relation of type SUBORDINATE_OF from 'stas khristov' towards 'actresses'?

ChatGPT

The sentence only mentions that Stas Khristov is a concert director, but **it doesn't specify whom he works for** or if he is subordinate to anyone.

Улучшение формирования "подсказок"

Подход, используемый в настоящем докладе носит название: Zero-shot обучение, т.е. отсутствие какой-либо вспомогательной информации (кроме контекста). Однако, могут быть рассмотрены следующие подходы для улучшения результатов:

- Few-Shot Обучение;
- Самоанализ рассуждений;

Самоанализ рассуждений

Устранение галлюцинаций сети: задание вопроса проанализировать результат рассуждений проведенных в рамках предыдущего ответа(-ов)⁵:

[Основной текст вопроса]. + **Let's work this out step-by-step to be sure we have right answer.**

Далее, можно спросить проанализировать результат, что практически может улучшить результат:

Check the correctness of the result for the provided step-by-step analysis.

⁵ <https://www.youtube.com/watch?v=wVzuvf9D9BU>

Собирая все подходы воедино

Изначально используемый формат подсказки:

For text "... " is the relation of type X from OBJ_A to OBJ_B ?

Сначала анализ, а затем вывод результата⁶

[Предварительный контекст, если требуется]. For text "... " is the relation of type X from OBJ_A to OBJ_B ? **Answer the question step-by-step and infer one of the result: "yes" or "no".**

⁶ Предположение основано на особенности архитектуры GPT-декодера и последовательной генерации текста ответа

Демо и используемые средства

```
python3 -m arekit_ss.sample --writer csv \  
--source nerel --sampler prompt \  
--prompt "For text: '{text}', is the relation of type \  
{label_val} from '{s_val}' towards '{t_val}'? \  
Answer yes or no, and explain why if no." \  
--src_lang ru --dest_lang en --text_parser lm --output_dir \  
./_out/nerel-prompting-fact-checking/ --splits train
```



LLM и графы



Способы применения и вывод

- ChatGPT может быть использована для Fact-Checking, оценка источников и ресурсов.
- Перевод запросов на английский хорошая практика (упрощенная семантика текста).
- изменение проекта на Codalab для адаптации к своим данным.

Спасибо за внимание!



<https://nicolay-r.github.io>